Daniel James Loza

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Education

Georgia Institute of Technology | Atlanta, GA *May 2029*
Bachelor of Science in Electrical Engineering, GPA 4.00

Skills

Programming and Tools: Electrical Design Automation (Cadence, Synopsys), Python/Data Analysis, SystemVerilog, Java, Git, Linux
Hardware: Photolithography Machines, Microcontrollers, PCBs, analog/digital circuitry, sensors, displays, soldering
CAD: Printed Circuit Board Design (KiCad), 3D Modeling (SolidWorks)
Communication: Presentation, research, team projects, and customer service experience
Certificates: Semiconductor Fabrication 101, Purdue University *August 2026*

Activities

Nanotechnology and Photonics Society | Project Lead *July 2026 – Present*
- Photolithography project technical lead; designing, billing materials, and assembling photolithography machines.
- Fabricating parts and electronics to make semiconductor device fabrication more accessible to GT students.
- Performing administrative tasks, including reviewing new member onboarding projects, buying materials, and managing teams.

SiliconJackets | Physical Design Team Member *October 2026 – Present*
- Designed, verified, and routed a partial RISC-V processor using SystemVerilog, Cadence software and Synopsys software.

Math Modeling | Team Member *2024 – 2026*
- Coordinated with a team to author 4 data-driven papers modeling and quantifying real-world problems.
- Applied mathematical modeling and computational analysis to propose, analyze, and present solutions.
- Won a $6000 prize from the Pennsylvania DOT for innovating a solution which reduces work zone-related fatal car accidents.
- Recognized by 2 international competitions, including placing in the top 20 out of 794 teams in the M3 Challenge.

Experience

Intellimation LLC | Norristown, PA *May – June 2026*
Electrical Engineering Intern
- Wired and programmed HVAC controllers to react to various room conditions and to control actuators.
- Programmed prototypes to remotely control vents and dampers, ensuring comfortable temperatures and gas levels.
- Shadowed and assisted field engineers to troubleshoot failing BACnet communications between buildings.

Hospital of University of Pennsylvania | Philadelphia, PA *June – August 2025*
Research Intern
- To predict patient hypertension outcomes after liver transplantations, analyzed and manipulated 900,000 patient records using mathematical models and Python, achieving a reproduceable 94% prediction accuracy.
- Co-authored pending research on health complications related to liver transplants, writing 9 analysis programs.

Cup of Dreams Coffee and Tea | Paoli, PA *2022 – March 2026*
Associate Manager
- Led shifts, managed scheduling and training for 10+ employees, served customers under pressure, and accounted for cash.

Projects

Cello Alarm | Personal Project *Summer 2026*
- Manufactured a remote-controlled motion-activated alarm using a printed circuit board to deter the tampering of valuables.
- Prototyped and tested software, I2C communication, and hardware to ensure reliable detection and triggering of the alarm.
- Utilized a ProtoMat and ProtoLaser to build the 2-layer PCB, integrating an Arduino, accelerometer, IR sensors, and buzzers.